Presenti.AI

SQL & Data Engineering Fundamentals

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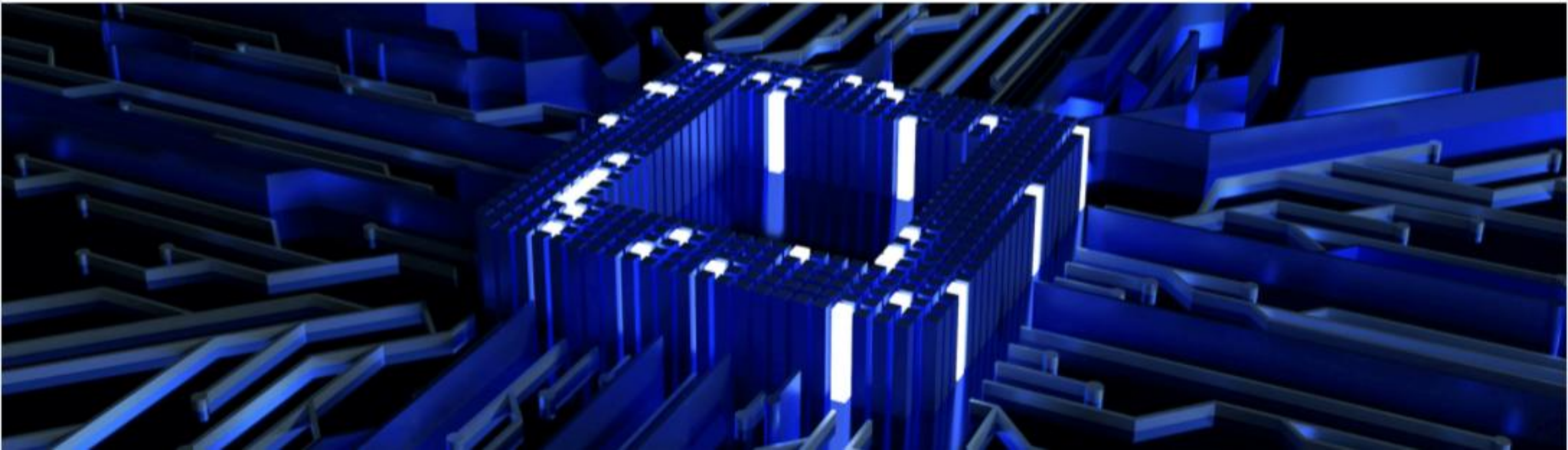
Section 1: Data Foundations & Core Concepts



What is Data?

1 Core Definition

Raw facts and figures collected from various sources become valuable when processed and analyzed for business insights.

2 Raw Data Limitations

Raw data has limited value; transformation and analysis unlock actionable business insights and competitive advantages.

3 Data Ubiquity

Data exists everywhere: transactions, user behavior, sensor readings, and social media interactions continuously generated daily.

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4 Real-World Application

Zomato collects order data revealing customer preferences and demand patterns, driving strategic business decisions and recommendations.

5 Foundational Importance

Understanding data fundamentals is essential for anyone pursuing data engineering and analytics careers and roles.

6 Business Impact

Data drives modern decisions: inventory management, personalized recommendations, and operational efficiency across all organizational levels.

Types of Data

01

Structured Data

Organized in fixed-schema tables like relational databases and spreadsheets. Customer records in banking systems .

02

Semi-Structured Data

Possesses organizational properties with flexible formats including JSON, XML, and logs. API responses and clickstream data demonstrate this type.

03

Unstructured Data

Lacks predefined format encompassing images, videos, and text documents. Social media posts and customer reviews exemplify unstructured data.

04

Amazon's Approach

Combines structured sales transactions, semi-structured product metadata, and unstructured customer reviews for personalized recommendation engine functionality.

05

Banking Industry Application

Integrates structured account data, semi-structured transaction logs, and unstructured loan documents for comprehensive risk assessment processes.

06

Data Volume Reality

Unstructured data comprises approximately eighty percent of enterprise data but demands specialized processing tools and infrastructure solutions.

Introduction to Databases

Definition & Core Concept

An organized collection of structured data stored and accessed electronically through a database management system for efficient management.

Real-World Analogy

Like a library system: tables are sections, rows are books, columns are attributes such as title, author, ISBN.

Primary Purpose

Enables efficient storage, retrieval, updating, and deletion of data through CRUD operations for optimal data management.

Key Benefit

Multiple users simultaneously access and modify data while maintaining data integrity, security, and system consistency standards.

Practical Example

Swiggy uses interconnected database tables storing restaurants, menus, orders, users, and delivery information seamlessly together.

Strategic Importance

Databases form the backbone of all modern applications, from mobile apps to large-scale enterprise systems worldwide.

02

Section 2: Database Architecture & Data Storage

Core Database Components

Fundamental Building Blocks

Tables organize related data entries structured like spreadsheets with rows and columns for efficient data management and retrieval.

Structural Blueprint

Schema establishes table structure defining column names, data types, constraints, and relationships governing database organization and integrity.

Practical Implementation

Zomato Order Table demonstrates schema with order_id Primary Key, customer_id, restaurant_id, order_date, total_amount, and status fields.

Records and Fields

Rows represent single entity instances such as individual customer orders while columns define specific properties like dates and amounts.

Unique Identification

Primary Keys uniquely identify each row preventing duplicates ensuring data integrity and enabling accurate record references across database operations.

Keys & Relationships

Primary Key (PK)

Uniquely identifies each record in a table; cannot be NULL; ensures data integrity and prevents duplicate entries.

Foreign Key (FK)

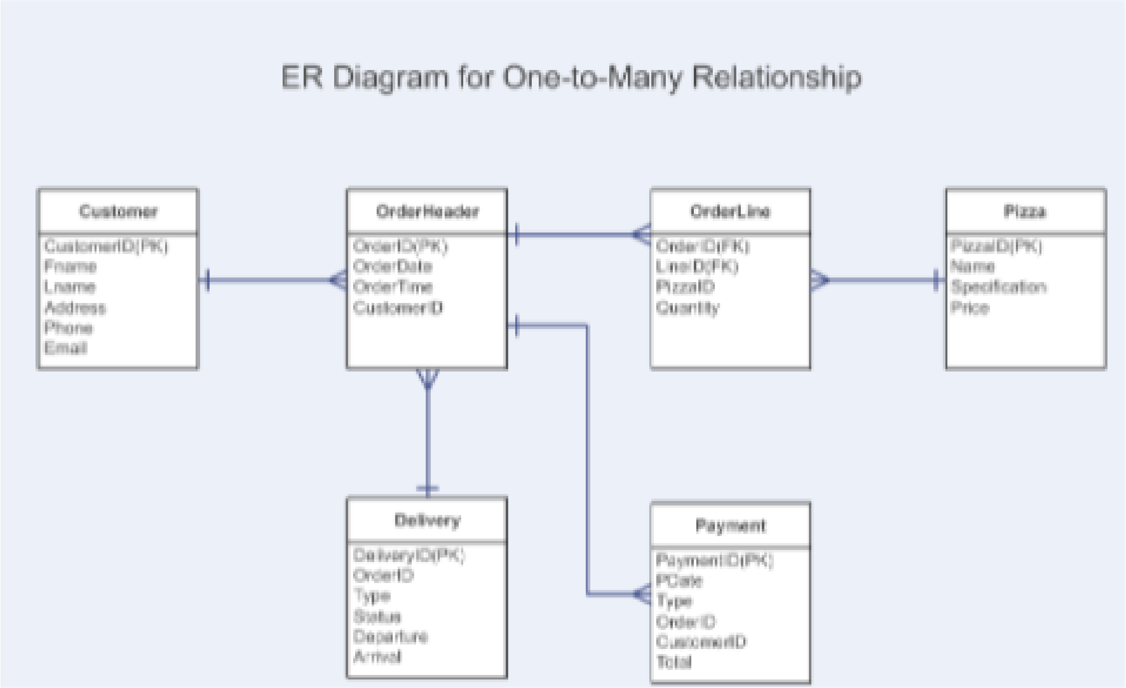
References primary key from another table; establishes relationships between tables for data consistency.

Composite Key

Combination of two or more columns that uniquely identifies a record within a table structure.

Relationship Types

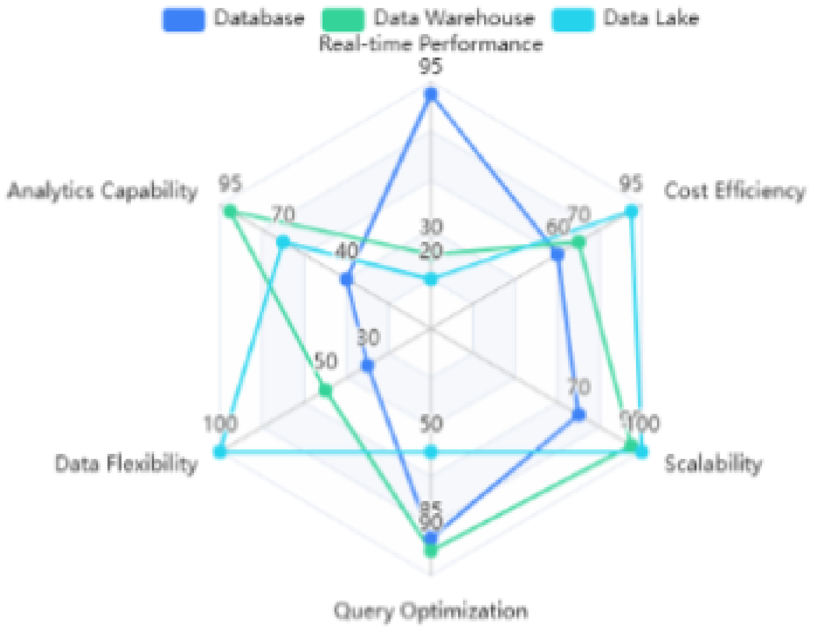
One-to-One (1:1), One-to-Many (1:N), and Many-to-Many (M:N) establish how database tables connect.



Real-World Case Study

Food delivery apps use FKs to connect Users → Orders → Restaurants → Dishes ensuring referential integrity.

Data Storage Systems Comparison



Database vs Warehouse vs Lake

Databases optimize OLTP with real-time transaction processing; warehouses enable OLAP for analytical queries; data lakes store raw unstructured data at scale for exploration and ML initiatives.

Architectural Integration Strategy

Modern data architectures leverage databases for operational efficiency, warehouses for business intelligence reporting, and lakes for data science exploration, creating comprehensive data ecosystems.

Characteristic	Database	Data	Data Lake
Use Case	OLTP Operations	OLAP Analytics	Exploration & ML
Cost	Medium	High	Low
Data Format	Normalized	Denormalized	Raw/Unstructured
Examples	PostgreSQL, MySQL	Snowflake	Hadoop, S3
Optimization	Row-oriented	Column-oriented	Schema-on-read

Modern Data Systems

Data Warehouse

- A **Data Warehouse** is a centralised system used to store **structured, cleaned, and historical data** for reporting and analytics.
- Stores **processed data**
- Optimized for **analytics (OLAP)**
- Used by **BI tools (Power BI, Tableau)**
- Ex. A company stores **5 years of sales data** to analyze revenue trends.

Data Mart

- A **Data Mart** is a **subset of a data warehouse**, designed for a specific business team.
- Smaller and faster
- Easier for teams to use
- Ex. Sales team uses only **sales data mart** instead of full warehouse.

Modern Data Systems

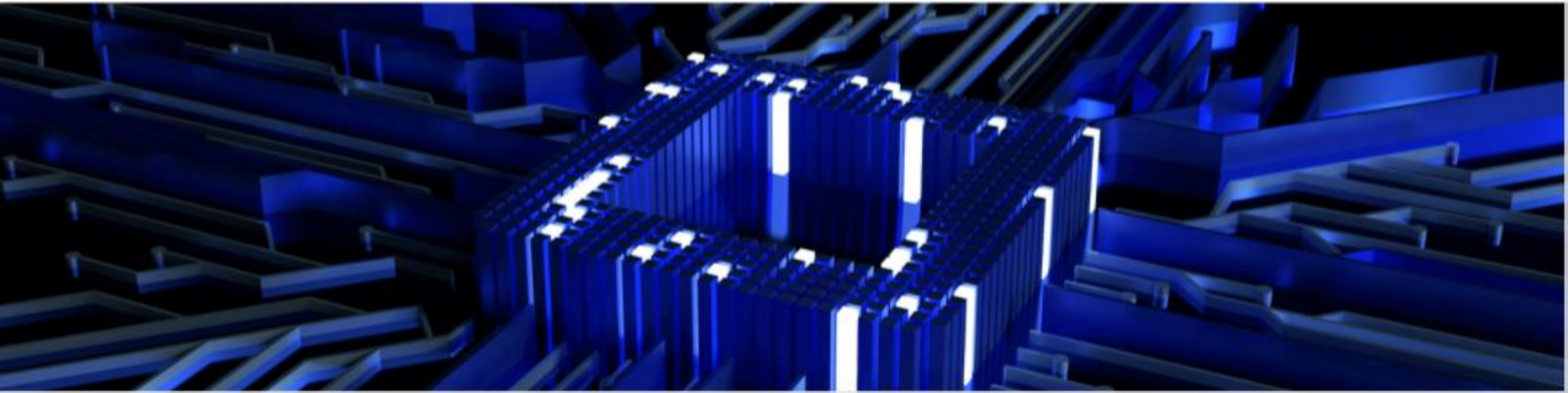
Data Lake

- A **Data Lake** stores **raw data in its original format** (structured, semi-structured, unstructured).
- Stores **everything as-is**
- Cheap storage (S3, ADLS)
- No strict schema

Lakehouse

- A **Lakehouse** combines the power of:
  - Data Lake (cheap storage, raw data)
  - Data Warehouse (performance, analytics)
- Supports **ACID transactions**
- Handles **all data types**
- BI tools can directly query it

03 Section 3: Data Modeling & Engineering Patterns



Data Modeling Concepts

Fact Table

Contains measurable, quantitative data metrics; stores business events like orders, clicks, and transactions; connected to dimensions via foreign keys.

Dimension Table

Contains descriptive attributes defining who, what, where, when; provides essential context for comprehensive analysis and reporting activities.

Star Schema

Fact table centered with radiating dimension tables; optimized for query performance; enables easy understanding and simple joins for BI reporting.

Snowflake Schema

Normalized dimensions broken into sub-dimensions; saves storage space but requires more complex queries; ideal for specific analysis scenarios.

Example - Zomato Analytics

OrderFact contains order_id, amount, delivery_time; connected to RestaurantDim, CustomerDim, DateDim, LocationDim for comprehensive analytics.

Modeling Decision

Choose star schema for speed and simplicity; select snowflake for storage optimization based on query complexity and data volume requirements.

ETL vs ELT Paradigms

2020
01

Traditional ETL Approach

Extract raw data, transform in staging area, load to warehouse. Safer, slower processing for sensitive data governance requirements.

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02

Modern ELT Approach

Extract, load raw data to lake, transform on-demand. Faster, flexible, cost-effective for cloud-native architectures.

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ETL Use Cases

Deploy for sensitive data, complex business logic, strict governance needs, and small-to-medium data volumes requiring safety.

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04

ELT Use Cases

Deploy for cloud platforms, big data exploration, iterative analytics, AI/ML workloads requiring raw data accessibility.

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05

Industry Shift & Hybrid Strategy

Cloud platforms enable ELT through compute separation. Enterprises adopt hybrid: ETL for sensitive ops, ELT for exploration.

Data Pipelines & Processing Modes

Data Pipeline Architecture

Orchestrated workflow moving data from source through ingestion, transformation, storage to consumption with automated scheduling capabilities.

Pipeline Components Flow

Data Source → Ingestion Layer → Transformation Engine → Storage Layer → Analytics/BI Tools sequential progression.

Batch Processing Mode

Processes large data volumes at scheduled intervals cost-effectively; acceptable latency measured in hours or days using Spark, Hadoop, DBT.

Hybrid Processing Approach

Combines real-time processing for critical metrics with batch processing for bulk analysis; exemplified by ride-sharing applications.

Monitoring & Orchestration

Tools like Airflow and Prefect ensure reliable pipeline execution, handle failures, maintain data quality SLAs systematically.

Real-Time Processing Mode

Continuous stream processing with milliseconds/seconds latency; higher cost; powered by Kafka, Flink, Spark Streaming technologies.

Industry Tools & Platforms

Snowflake

Cloud data warehouse with SQL native support, auto-scaling capabilities, and zero-copy data sharing for enterprise analytics.

Databricks

Unified analytics platform built on Apache Spark with Delta Lake integration for data science and analytics workflows.

AWS S3

Petabyte-scale object storage for data lakes, cost-effective raw data management, and seamless AWS tool integration.

Power BI / Tableau

Business intelligence platforms enabling self-service analytics, interactive dashboards, and end-user reporting capabilities.

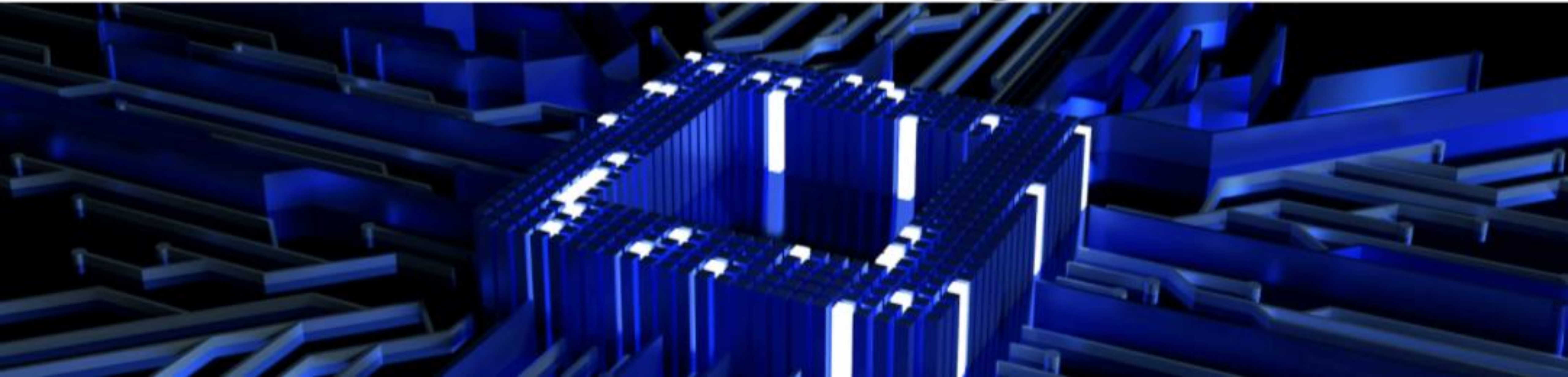
Apache Spark

Distributed processing engine supporting batch and streaming with Python, SQL, and Scala APIs for large-scale transformations.

Technology Stack Integration

Raw data flows from S3 to Databricks, transformed via Spark/SQL, loaded to Snowflake, and visualized in Power BI.

04 Section 4: Real-World Application & Learning Path



Real-World Case Study: Food Delivery App

01

Stage 1 - Data Collection

App captures orders, delivery tracking, GPS coordinates, customer reviews, and payments in JSON logs format stored in AWS S3.

02

Stage 2 - Data Processing & Cleaning

Deduplicate orders, handle missing values, standardize formats, and create surrogate keys using Spark SQL, Python, and dbt tools.

03

Stage 3 - Analytics-Ready Warehouse

Build fact and dimension tables using star schema optimization, stored in Snowflake or Databricks for efficient query performance.

04

Stage 4 - Business Intelligence

Create dashboards tracking daily orders, restaurant performance, delivery efficiency for business analysts, restaurant partners, operations team.

05

Stage 5 - Advanced Analytics

Deploy ML models predicting demand, delivery time, customer lifetime value to optimize inventory, staffing, and personalization strategies.

Data Engineering: Core Concepts & Career Path

1 Core Takeaway

Data engineering systematically transforms raw data into business insights through storage, transformation, and delivery architectures for organizational value.

2 Architecture Evolution

Evolution progresses from rigid relational databases to flexible data lakes, culminating in intelligent lakehouses that support comprehensive analytical workloads.

3 Key Concepts Review

Databases organize structured data; warehouses optimize analysis; lakes provide scalability; lakehouses intelligently combine flexibility with analytical performance.

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4 Modern Best Practices

Adopt cloud platforms for scalability; implement medallion architecture for data quality; separate storage from compute; establish robust data governance frameworks.

5 Future Readiness

AI automation handles routine tasks; foundational data knowledge ensures career longevity, advancement, and relevance in evolving technology landscape.

6 Hands-On Learning Path

Master SQL fundamentals, practice datasets progressively, transition to cloud platforms, and actively contribute to production data pipelines.

Thanks